

CHS Handover Notes

CHS — Handover Notes

Compiled: August 25, 2026, last updated September 10, 2026 — covering the full engagement to date, kept current alongside every batch of work, the same discipline already applied to the Users Guide, Terms & Conditions, and Feature Catalog.

Companion document: CHS_PROGRESS_LOG.md — a chronological record of what was built, in order. Read this file first for orientation, then the log for the detailed sequence.

1. What CHS Is

Complete Housing Solutions (CHS) — a Nigerian real estate PWA covering the full property journey: buying, renting, leasing, shortlet booking, and property sales, currently focused on Kaduna State. CHS is a genuine facilitator, not a passive listings board — it verifies documents and identities, holds funds in real escrow, and runs its own dispute-resolution process.

2. Stack & Repository

- **Frontend:** Next.js 16 (App Router), TypeScript, Tailwind CSS — repo folder chs-v2/
- **Backend:** Supabase (Postgres + Auth + Storage + Edge Functions), project ref havwhdgjqgtxtqklkqfm
- **Payments:** Paystack — currently on a **test key** (pk_test_...), not yet switched to live
- **Hosting:** Netlify, connected to GitHub for auto-deploy (confirm this is still true before assuming a push auto-deploys — it was manual drag-and-drop earlier in the project's history)
- **Migrations:** backend/ (01-10, original schema) and backend-v2/ (11-105, everything since) — sequential, numbered, meant to be read and applied in order
- **Documentation:** CHS_USERS_GUIDE.docx (+ in-app /guide), CHS_COMPLETE_FEATURE_CATALOG.pdf (+ in-app /admin/feature-catalog, + segmented CHS_FEATURE_CATALOG_SEGMENTED.xlsx), this handover note, and the progress log

3. Real Architectural Patterns Established

A new agent should follow these, not reinvent them:

- **RLS via staff_can_access(domain)** — every admin-facing table checks a real staff domain (finance, customer_care, registration_setup, owner_buyer_tenant, agent_relations, artisan_dev_pm_vendor), not a blanket “is admin” check. finance is deliberately not assignable to any sub-admin — Super Admin only, by design.
- **High-stakes actions go through a real approval queue**, not a direct write. See request_admin_action() / apply_admin_action() / resolve_admin_action() in backend-v2/51_subadmin_roles_and_approval_queue.sql onward. Anything

financial or verification-related must route through this, not a raw table update.

- **Test data is always explicitly marked**, never silently mixed with real data. Pattern: `is_test_grant` boolean columns, `TEST-` prefixed references, and description text starting with `△ TEST FUNDING`. Follow this pattern for any future test data.
- **Real, not simulated, verification.** Bank accounts are checked against Paystack's real account resolution API before they can be linked (resolve-bank-account Edge Function) — never trust a self-typed account name again.
- **Money-moving Edge Functions never trust client-supplied amounts.** Real fees are read server-side from `platform_settings`, not passed from the browser.
- **Apply every migration directly to the live database as it's built**, and test with real data (using `set role authenticated + set_config('request.jwt.claims', ...)` to simulate a real user's RLS context) before considering it done. Supabase is always current; GitHub/Netlify only reflect the latest zip once actually pushed — these are two separate things on two separate schedules.
- **Always save every migration as a real local file in backend-v2/, the same moment it's applied to the live database — never apply-only.** A real, serious gap was found and fixed: migrations 77 through 95 were applied and tested live but never saved as files, meaning the exportable project and the live database quietly drifted apart for several rounds of work. Verify the file count directly inside a packaged zip before calling a package complete — don't assume the fix worked.
- **An “approve/confirm/mark paid” action must trigger its real consequence, not just update a status.** A comprehensive audit found 7 separate real instances of the same bug: a status changed and a notification fired, but no money, role, or state actually moved. Whenever building or reviewing any approval-style action, explicitly ask “what real thing is supposed to happen now, and does this code actually do it” — then prove it with a real before/after balance or state check, not just a read of the code.
- **When copying a “finished” documentation file between folders, always copy FROM the just-edited version TO the other location — never the reverse.** A real mistake happened doing exactly this backwards: freshly-updated Progress Log and Handover Notes were overwritten by their own stale originals from a different folder. Verify file content after any cross-folder copy, not just that the copy command succeeded.

4. Known Gaps — Read Before Assuming Something Works

Fixed since the last version of this document (kept here briefly so a new agent doesn't waste time re-discovering these): identity verification self-certified without real admin review; mobile logout was only reachable from the homepage; House Rules appeared on Sale listings where it made no sense; the “similar properties” advertising system had a real geographic-matching bug; sellers had no way to attach a real message when responding to an offer; the rent countdown and 90-day non-renewal notice had never been built despite the real database column existing; **most significantly, there was no real mechanism anywhere for a buyer to actually pay a seller the purchase price of a property** — only commission had ever been tested. All of these are detailed in `CHS_PROGRESS_LOG.md`.

Still genuinely open:

- **No real recurring billing exists for any promotion package** (Classic/Premium/Elite/Signature), including Signature specifically. The database rows and pricing exist; the actual Paystack

subscription charge was never built. Anyone assigned one of these packages today is effectively getting it free.

- **Pre-launch admin test mode** (Admin → Overview → “Switch role for testing”) and the **Construction Roadmap test-unlock button** are both explicitly temporary. Remove both — and the underlying `grant_roadmap_access_test()` function — before real launch.
- **A real, substantial set of demo accounts exist in the live database**, built specifically so the client could manually test negotiation → payment → commission across categories: phone numbers 08050000001 through 08050000010, PIN 123456 for all, covering Land/House/Office/Hotel/Shortlet buyer-seller and host-guest pairs, with real funded wallets and real live listings. These are clearly marked by name (Demo Land Seller, etc.) but are real, fully-verified, functioning accounts — clean up or keep, but know they exist before launch.
- **The mobile viewport/horizontal-scroll bug has been fixed twice** (a missing viewport meta tag, then a root-level overflow-x: hidden after the first fix reduced but didn’t fully eliminate it) — **awaiting the client’s next real-device confirmation** that it’s genuinely resolved. Check with them before assuming this is closed.
- **True server-side pagination was never built.** The homepage and admin queues are capped (e.g. `.limit(300)`), not genuinely paginated — fine for now, will need real work as the catalog grows.
- **Paystack is still on a test key.** Real payments do not move real money yet.
- **No custom domain configured.** Still on the default Netlify subdomain (extraordinary-conkies-312c3d.netlify.app) — note that WebAuthn’s `WEBAUTHN_RP_ID`/`WEBAUTHN_ORIGIN` secrets will need updating to match whenever a real domain is set up, since WebAuthn ties strictly to the exact origin.
- **WebAuthn’s domain configuration could not be directly verified.** The cryptographic implementation itself is genuinely solid, but secret values can never be read back once set — only overwritten. If biometric login ever seems to fail for every real user, this is the first place to check.
- **Real architectural design generation does not exist**, and was explicitly confirmed not to be a realistic near-term build — it would require either specialized generative-design/CAD software or real, licensed architects, not standard web development. Engage CHS’s document delivery is a real human-upload tracker, not a generator.
- **The real sale-payment and escrow-hold mechanism is new and has only been tested with synthetic data, not a real end-to-end client walkthrough yet.** Genuinely correct in every test run, but worth a close first real-world pass — particularly the interaction between required-document verification and the payment gate.
- **Database backups / point-in-time recovery on Supabase — never confirmed.**
- **No uptime or error monitoring** — an outage today would be discovered from a user complaint, not an alert.
- **13 of the original 16 app views were never systematically compared against the original design spec** (a note from the very first handover, still open).

5. Major Systems Built Since the Last Version of This Document

- **Reminder Engine** (backend-v2/69) — one real, generic escalation system: rent-due cadence, maintenance follow-up escalation, and shortlet guest-message escalation, all sharing the same infrastructure. Multi-channel (in-app always works; SMS via Termii and email via Resend depend on real provider secrets being

configured).

- **Estate Management** (backend-v2/70, 77, 78) — Estates as a real entity distinct from individual property delegation, with real service charges, bulk CSV unit onboarding, and a real subscription model (₦20k/50k/110k monthly by unit-count tier, confirmed with the client).
- **The real, complete commission model** (backend-v2/75, 76, 91) — Sale (6.5%/6%), Rental (5%/5.5%), Shortlet (real sliding scale by length of stay), Hire/Booking flat tier (Hotel/Event Centre/Car park), and Rent to Own/Mortgage (5%/5.5% per installment) — see the actual Terms & Conditions text (components/TermsContent.tsx) for the definitive, current numbers, since this is the one place they must never drift out of sync.
- **Rent to Own / Mortgage** (backend-v2/91, 94) — a genuinely new listing category, not the same thing as “Hire.” Real installment payments, real ownership-percentage tracking, automatic conversion to a completed sale at 100%.
- **Shortlet/Hotel guest-host messaging** (backend-v2/92) — real in-app communication with genuine host anonymity toward the guest, never toward CHS itself.
- **Maintenance Reserve, now genuinely functional** (backend-v2/95) — real external funding, real artisan-payment draw order (reserve first, main wallet for any shortfall), real unrestricted withdrawal.
- **The real Sale purchase-price payment mechanism** (backend-v2/101, 102) — the single most significant fix of the engagement. Previously, only commission had ever been built or tested for a Sale; there was no way for a buyer to actually pay a seller. Now a single, transparent checkout: buyer’s total automatically includes their own commission (shown as both percentage and real value), seller’s net automatically excludes theirs.
- **Real, sourced legal document requirements and an escrow hold for Sale proceeds** (backend-v2/103, 104, 105) — Certificate of Occupancy, Deed of Assignment, Survey Plan, Governor’s Consent, Tax Clearance Certificate, Sale Agreement, and conditionally Building Plan Approval, sourced from genuine Nigerian legal references. A buyer’s payment is blocked until every required document is CHS-verified; a seller’s proceeds are held (visible, not withdrawable) until CHS confirms the real, physical legal transfer is complete.
- **Real, admin-reviewed buyer identity verification** (backend-v2/96-98) — previously self-certified the instant the form was filled in; now a genuine pending → admin-review → approved flow.
- **Rent countdown and 90-day non-renewal notice** (backend-v2/99) — a real, live day-count on the tenant dashboard and a genuine notice mechanism, built after confirming the underlying database column had existed unused since early in the project.

New Since Last Handover Update (migrations 107-191)

- **Agent/Manager business tools** (145-149, 154-158, 175) — a real team/staff system (invite, role labels, daily reports), tenant and property/owner registers, expense tracking, owner remittance, auto-generated digital receipts, an independent per-owner commission model, and multiple bank accounts.
- **Host and Guest as real, separate roles** (160, 178-182) — split from Owner/Buyer with their own dashboards. The entire shortlet/hire payment mechanism was rebuilt: real request-to-book (guest sees full cost including commission before committing), a genuine host accept/decline, automatic refund on decline, and admin-released host payout — previously, hosts had no way to ever get paid at all.
- **Ratings, cancellation, and security deposits** (183-186) — real two-way ratings after a confirmed stay; a stated, server-enforced

cancellation policy (100%/50%/0% by time-to-check-in); a flexible, host-toggled security deposit that auto-waives after a guest's 3rd real rating, resolved by admin (release or claim) rather than left to either party's word.

- **Developer as a real, separate role** — previously registerable but with no login path and no dashboard at all; now built the same way as Host/Guest.
- **Three real KYC bugs found and fixed:** no field existed to enter a non-NIN ID's actual number (the system silently substituted NIN regardless); admin's document review only checked a person's *primary* role, missing real credentials on a secondary role; and stored, pre-signed document URLs are structurally fragile, now replaced with a fresh signed URL generated live at the moment of viewing.
- **Property reference numbers and admin search** (187, 188) — every property now has a permanent PROP-000001-style reference, backfilled for all existing properties, with a real admin search by reference/title/owner. Also fixed: an application awaiting an owner's decision was previously invisible to admin's own query.
- **Real PIN reset** (189) — phone + NIN verification; did not exist in any form before this.
- **Real FAQ page** — written from CHS's own T&C and actual features, reachable from the previously-placeholder "Help & FAQs" menu entry.
- **Add-a-role without logging out, plus a role switcher** — Profile page now shows every real role as a tappable pill and links straight to "Add another role," which skips the phone/PIN re-verification entirely for an already-authenticated session.
- **Desktop splash screen fix and four new per-role color themes** (Host, Guest, Developer, Staff) — a real, deliberate 7-zone color system already existed; these four roles were quietly borrowing another role's colors and now have their own.
- **Team/Staff subscription revenue model** (190, 191) — free for 2 staff, real subscription from the 3rd, tiered by staff count, with exact multi-month discount math (6 months paid → 8 months access; 12 paid → 18 access).

Supabase Status (confirmed at time of writing)

Every migration through 222_enable_realtime_notifications is live in the real, production Supabase project — checked directly against Supabase's own migration history, not assumed from local files. No drift between the exportable backend-v2/ folder and the live database. One real ERROR-level security advisory is outstanding (a SECURITY DEFINER view, public.public_profiles) and worth a follow-up look; the remaining advisories are standard hardening warnings, not active vulnerabilities.

New Since Last Handover Update (migrations 192-222)

- **Receipt/voucher redesign** — the real, live receipt page rebuilt with a genuine branded design (gradient header, watermark seal, serif amount display); no schema change, frontend only.
- **Artisan commission corrected** (201) — real, tiered 6%/8.5% by labor amount, replacing an earlier flat 10%.
get_artisan_commission_pct() is the real source of truth; the old flat setting in platform_settings is dead, harmless data.
- **Complete marketplace rebuild** (196-197, 202-206) — admin-mediated, contact-filtered messaging (reusing detect_offplatform_contact); real escrow-backed payment at 6%/4% (accept_marketplace_quote,

release_marketplace_escrow_to_vendor,
refund_marketplace_escrow_to_buyer); a real direct-buy path
(buy_product_direct) bypassing negotiation entirely. The old
referral_fee_settings/referral_fees_owed flat-fee tables are now
fully dead — nothing in current code reads from them.

- **Sister marketplace link** (207) — a real, admin-editable URL
(get_sister_marketplace_link, reusing the contact-settings pattern),
shown on Marketplace, Construction Roadmap, and Owner →
Maintenance. **Still holds a placeholder URL — must be
updated by admin to the real address.**
- **Rental application rework** (208) — real applicant_full_name,
employer_business_name, employer_business_address columns;
owner's review card rebuilt to show every real applicant field, not
just the guarantor's.
- **Independent guarantor verification** (211) — real, complete
redesign. submit_rental_application now takes only the guarantor's
name/phone; a real, single-use token (GTN-...) is generated and the
guarantor completes everything else about themselves at
/guarantor-confirm/[token], requiring no CHS account. A real
awaiting_guarantor_confirmation status genuinely blocks the
application from reaching the owner until this is done.
- **Event Centre real pricing** (212-214) — new event_capacity_tiers
and event_facilities tables, host-manageable at /event-
pricing/[propertyId]; request_event_booking computes and charges
the real total (tier + facilities + 6%) in one call. A real, verified
reference (the project's earlier prototype index.html) was checked
directly before rebuilding this. Simple event-day add-on flags
(band/caterer/ushers/notes) also added directly to
shortlet_bookings for the standard booking flow.
- **Real payment regressions found and fixed** (215-218) — pay_rent
and pay_rent_to_own_installment both silently omitted the payer's
own real commission from the actual amount charged; both
rewritten to compute and charge the true total in one atomic call,
and both return a real JSON breakdown (reference,
real_total_paid, etc.) the frontend now displays before payment.
- **Stale commission early warning** (219) —
get_stale_uncollected_commissions(), surfaced as a real, visible
alert on Admin → Finance for any commission invoiced but
genuinely uncollected 2+ hours — the concrete safeguard against
the above regression pattern recurring silently.
- **Clickable, real-time notifications** (220-222) —
notify_admins_by_domain widened to accept a real p_link (it never
did before); real links threaded through the full rental-application
decision chain; **the notifications table is now genuinely in
Supabase's real-time publication** (it never was before — this
was the actual, confirmed cause of "no admin pop-up").
NotificationBell.tsx now subscribes via postgres_changes and
shows a real, visible toast on arrival.
- **New real, dedicated pages:** /my-applications (buyer/tenant, all
applications+offers), /owner-applications (owner, all received
applications+offers, most recent first), Admin → Platform Earnings
tab (real payer/amount/timestamp). Rent Savings widget removed
from the general homepage — confirmed the tenant dashboard's
own WalletQuickView already covered it; was a genuine duplicate,
not a missing feature.

New Since Last Handover Update (migrations 223-274)

This was the single largest batch of the engagement — a long,
iterative round of real client testing that surfaced genuine bugs across
nearly every part of the app, plus several substantial new features
built in direct response to real, repeated client requests. Grouped by
theme rather than strict migration order, since many of these
interleaved:

- **Notification click-through — three real attempts, the third one correct.** The underlying problem was never “no link exists” — it was how the click was wired. Attempt one (a Link wrapping a clickable div) was structurally fragile. Attempt two (`window.location.href`) fixed the click but caused a genuine regression: a full, disruptive app reload that lost in-progress admin work. The real, final fix is `router.push()` combined with explicit `stopPropagation/preventDefault` on the click handler — a proper Next.js soft navigation, not a hard reload. `NotificationBell.tsx` is the one shared component; fixing it once covers every dashboard.
- **Admin mediation extended to rental applications and offers, both directions** (235, 238, 243) — closing a real, serious gap: a buyer’s offer, and separately an owner’s accept/decision on that offer, both used to reach the other party directly with zero CHS review. Real awaiting_admin_review / owner_decided_pending_relay statuses now gate both directions, matching the same real pattern the marketplace already used. `admin_relay_application_to_owner`, `admin_relay_offer_to_owner`, and `admin_relay_offer_decision_to_buyer` are the three real relay functions — all now audit-logged (see below).
- **A buyer’s real phone number, name, occupation, and source of funds** (233) — an offer previously required no real bio-data at all. Now required at submission, shown to admin for verification, and deliberately kept off the owner’s own screen — contact stays mediated through the real, contact-filtered message thread, the same principle already enforced in the marketplace.
- **Admin document review, the single most serious gap found this round** (244–251) — the screen where admin approves a new listing showed only the title and location, with zero visibility into the real uploaded legal documents, on a completely separate tab from where those documents could even be viewed. Fixed properly: every real document now shows directly on the approval card, with its own view link and verify/reject action, and — for Sale properties specifically — the “Verify” button is now genuinely disabled, not just a visual warning, until every document is verified and the owner’s own ID is confirmed verified too.
- **Land and other non-residential categories showing a bedroom count** — the display only ever checked whether a database value was non-null, never whether the category should have one at all. Fixed with a real, shared `shouldShowBedrooms()` safeguard, and cleaned 36 real listings that had this bad data.
- **Real room/bedroom/toilet dropdowns, and a genuinely repeatable “Others” facility field** — replacing plain number inputs; a new, separate toilets column (previously conflated with bathrooms); a new `total_rooms` field; the already-existing but never-wired `other_facilities_jsonb` column finally put to real use.
- **A genuinely serious “processed history” gap** — the real backend function for this existed from an earlier round but was never actually connected to any real admin screen; approved/rejected items simply vanished from view with no way to refer back. Extended to cover property listings and registrations (not just applications and offers), added real decision timestamps, and — critically — built the actual missing screen (`get_admin_processed_history()`, a new “Processed History” admin tab).
- **Audit Trail** (253–274, ongoing) — a real, permanent `audit_log` table with no update/delete permission for anyone, by design, plus a reusable `log_audit_event()` function. Wired into every real high-stakes action found so far: property verification/rejection, dispute rulings, escrow fund release, account suspension, all three admin relay functions, agent referral payouts, all three marketplace moderation actions, and agent management changes (link/revoke/replace). Genuinely not “every significant action” in the strictest sense yet — this is real, ongoing coverage, not a finished claim.
- **Feature Explainer** (ongoing) — a real, reusable `InfoTip`

component, mobile-first (tap opens/closes it; hover also works on desktop), live on 16+ real, previously-confusing terms across buyer, owner, and tenant screens (commission, escrow, guarantor, CHS Agent ID, rent auto-pay, the difference between move-in/move-out condition reports, and others). Real, working foundation; extending coverage across the rest of the app's 100+ features continues as ongoing work.

- **Property manager listing access** — a real, confirmed gap: agents could list a property, managers genuinely could not. Fixed to match the identical pattern already used for agents; no separate database-level restriction ever existed, this was a frontend role-gate that simply never included manager.
- **Real room-video system, a cost-free alternative to a paid virtual-tour service** (264-274) — new `property_videos` and `video_requests` tables. An owner/agent/manager records or uploads a short, real video per room, on both new listings and (added after an honestly-flagged gap) existing ones via `/edit-listing/[id]`. A buyer/tenant can request a specific additional video if unsatisfied, and can now track that request's real status on `/my-applications` rather than only getting a one-time confirmation. Found and fixed three further real gaps purely by exercising the feature end to end: the owner's pending-request view had no direct link to actually act on it; the real notification itself (the first thing an owner would tap) linked to the public property page instead of the real edit page; both fixed.
- **Document Site** — a new real admin page (`/admin/document-site`) bundling every current reference document (User's Guide, Terms & Conditions, Feature Catalog in two formats, this Handover Notes file, the Fix Tracker, the Property Manager Guide, and the System Documentation the client separately provided) directly into the app's `public/` folder, so they're always available without depending on a separate storage service.
- **New Round 2 demo accounts** (phone range 0812... , PIN 123456) — the original demo accounts were exhausted/mixed together across extensive testing. Five fresh accounts: an Owner (Land/House/Office/Warehouse listings), a Host (Chalet/Hotel/Event Centre, the Event Centre with real capacity tiers seeded), and Buyer/Tenant/Guest each funded at a real ₦100,000,000.
- **A genuine, real bug in how these Round 2 accounts were first created** — inserting directly into `auth.users` via raw SQL left several fields (`confirmation_token`, `recovery_token`, and related) as `NULL` where Supabase's real auth service expects an empty string, a documented pitfall of bypassing the official Admin API. Silently broke login for all five accounts. Found by a direct, full-field comparison against a known-working account; fixed by correcting the fields and re-hashing the password at the matching bcrypt cost factor.
- **A more saturated, "solid" color palette** — the client's core brand colors (charcoal + burnt amber) already matched a professionally-produced reference palette's own "best choice" option almost exactly; the real gap was that the individual per-role zone colors were noticeably more muted by comparison. All 14 zone colors upgraded to bolder tones drawn from the same reference.

Supabase Status (confirmed at time of writing, this update)

Every migration through 274 is live in the real, production Supabase project. No drift between the exportable `backend-v2/` folder and the live database. The same one real ERROR-level security advisory from the last update (`public.public_profiles`, a `SECURITY DEFINER` view) remains outstanding and still worth a follow-up look.

A note on process, worth reading before picking up this project

Several real mistakes happened this round, and each one was caught and corrected the same way: by actually testing against real data and the real, full build, not by trusting that a change compiled or looked right. A JSX edit that accidentally deleted a needed line; a React state-in-effect warning; a wrong TypeScript type for a Postgres join that returns an array, not a single object; a genuinely garbled shell heredoc that skipped packaging a zip entirely. None of these were caught by writing careful code the first time — they were caught by a disciplined habit of verifying afterward: full lint, full production build, direct database queries against real inserted test data, and — after one packaging mistake — actually re-opening the finished zip and searching it for the real code before calling anything done. Keep that habit. It's the actual reason “done” has held up more often than not.

6. Where to Find Things

- **Every real reference document in one tap:** Admin → 📄 Document Site (/admin/document-site) — User's Guide, Terms & Conditions, Feature Catalog (both formats), this Handover Notes file, the Fix Tracker, the Property Manager Guide, and System Documentation.
- **Every real feature, cross-referenced to its admin location:** CHS_COMPLETE_FEATURE_CATALOG.pdf, the segmented .xlsx, or in-app at /admin/feature-catalog (always current, same source data as the PDF).
- **How to use the app, by role:** CHS_USERS_GUIDE.docx or in-app /guide.
- **Resolving a specific user's support issue:** Admin → 🔍 Trace an Account — search by phone/email/name, see their full wallet, promotion, roadmap, and Engage CHS history in one place.
- **What a sub-admin did and when:** Admin → Overview → “View sub-admin action history.”

7. Standing Discipline Going Forward

Every future batch of work should: 1. Apply real migrations directly to the live Supabase database, and test with real data before considering it done. 2. Update CHS_PROGRESS_LOG.md with what was actually done. 3. Update this handover note if a new architectural pattern, gap, or critical fact emerges. 4. Update the Users Guide and Feature Catalog (both formats) if user-facing or admin-facing functionality changed. 5. Rebuild and verify (lint + full production build) before packaging anything for GitHub/Netlify. 6. **When testing a real function that's expected to fail, run the setup/funding step as a separate call from the failing call.** Supabase's implicit transaction batching means a later statement's error can silently roll back an earlier, successful update in the same request — caught more than once by verifying real database state after a test rather than trusting a query appeared to succeed. 7. **Before assuming a client-reported visual bug is fully fixed, get their explicit confirmation from a real device.** A compiled, built, and packaged fix is not the same as a confirmed fix for anything involving rendering — this class of bug has needed a second real pass before, and standing here waiting on that same confirmation again. 8. **When researching a real-world legal or regulatory requirement (documents, permits, compliance) for a feature, cite genuine, current sources and say so plainly — never invent a plausible-sounding list.** This mattered directly for the Sale legal-document requirements.